

Sensor-Based Municipal Water-Leak Monitoring

Project manager | Four-person academic team | University of Windsor | 2026

The project assessed a practical monitoring strategy for Ontario municipalities using pressure sensing, targeted acoustic monitoring, and a central data platform.

3	technology layers
4	team members
5	evaluation dimensions
Pilot	recommended first step

ROLE

Managed the project structure, schedule, task coordination, technical integration, and final communication across the four-person team.

ASSESSMENT

- Technical: detection capability, installation, data quality, and integration.
- Economic: capital and operating requirements and lifecycle value.
- Environmental: avoided water loss and reduced emergency repair impacts.
- Social and regulatory: service reliability, disruption, privacy, procurement, and municipal requirements.

RECOMMENDED ARCHITECTURE

Use pressure monitoring as the network-wide signal, deploy acoustic tools in targeted high-risk areas, and consolidate alerts and records through a central municipal data platform.

IMPLEMENTATION DECISION

Begin with a measured pilot in high-risk zones. Validate detection performance, false alarms, maintenance effort, and lifecycle value before broader deployment.

LIMITATIONS

The study was a technical-communication assessment, not a detailed design. Device selection, hydraulic modelling, cybersecurity, procurement, and field calibration require project-specific engineering work.

Evidence basis: team solution report, implementation schedule, presentation, and technical literature review. This is an academic strategy summary, not a municipal tender or issued-for-construction design.